

pfsense Firewall Installation and Network Configuration

1.Download pfSense ISO

1. Download pfSense CE ISO.
2. Create a bootable USB.
3. Boot the physical server from the USB.

2. Install pfSense

1. Select **Install pfSense**.
2. Accept the license agreement.
3. Choose **Auto (ZFS)** or **Auto (UFS)** installation.
4. Select the target disk.
5. Wait until installation completes.
6. Reboot the system.

3. Configure Interfaces

Assign interfaces:

- WAN → Connected to Internet
- LAN → Connected to Internal Network

Example:

WAN : em0

LAN : em1

4. Configure LAN IP Address

Example:

IP Address : 172.18.0.1

Subnet Mask : /20

DHCP Range : 172.18.0.100 - 172.18.15.254

5. Enable DHCP Server

Navigate to:

Services → DHCP Server

Configure:

- Enable DHCP Server
- Start IP : 172.18.0.100
- End IP : 172.18.15.254
- DNS Server : 8.8.8.8

Save and Apply.

6. Configure Firewall Rules

Navigate to:

Firewall → Rules → LAN

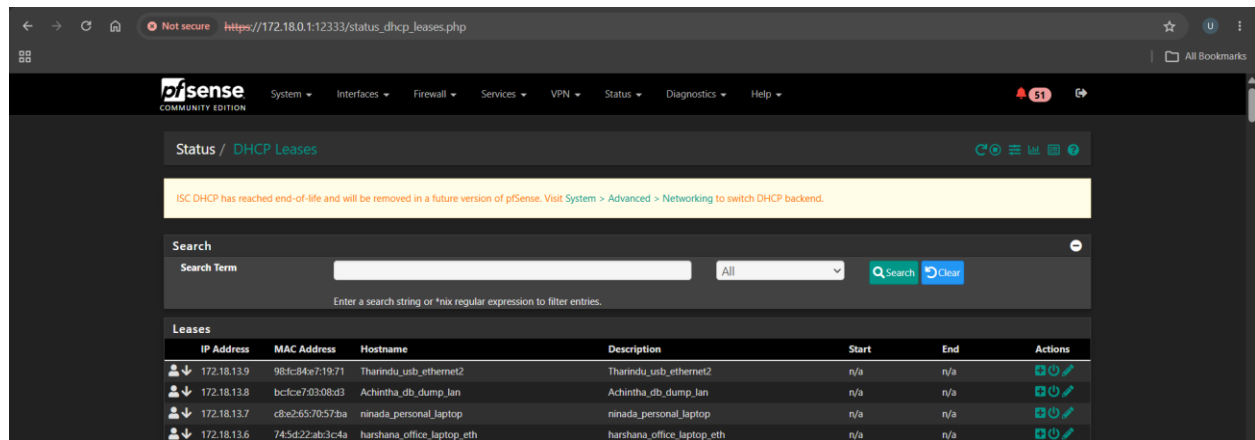
Allow:

- ICMP (Ping)
- HTTP (80)
- HTTPS (443)
- DNS (53)
- SSH (22)

7. Access pfSense Web Interface

Open browser:

<https://172.18.0.1>



8.Login to pfSense

Open:

`https://172.18.0.1`

Login with:

Username: admin

Password: *****

9.Open DHCP Server

Navigate to:

```
Services
├── DHCP Server
│   └── LAN
```

Scroll down to:

DHCP Static Mappings for this Interface

Click:

+ Add

10.Add Proxmox Server

Fill:

MAC Address : BC:24:11:AA:BB:01
IP Address : 172.18.8.5
Hostname : radius-server
Description : proxmox

Save → Apply Changes

11.Add FreeRADIUS Server

Fill:

MAC Address : BC:24:11:AA:BB:01
IP Address : 172.18.8.16
Hostname : radius-server
Description : Ubuntu VM - FreeRADIUS + daloRADIUS

Save → Apply Changes

12.Add Snipe-IT Server

Again click + **Add**

Fill:

MAC Address : BC:24:11:AA:BB:02
IP Address : 172.18.8.15
Hostname : snipeit-server
Description : Ubuntu VM - Docker + Snipe-IT

Save → Apply Changes